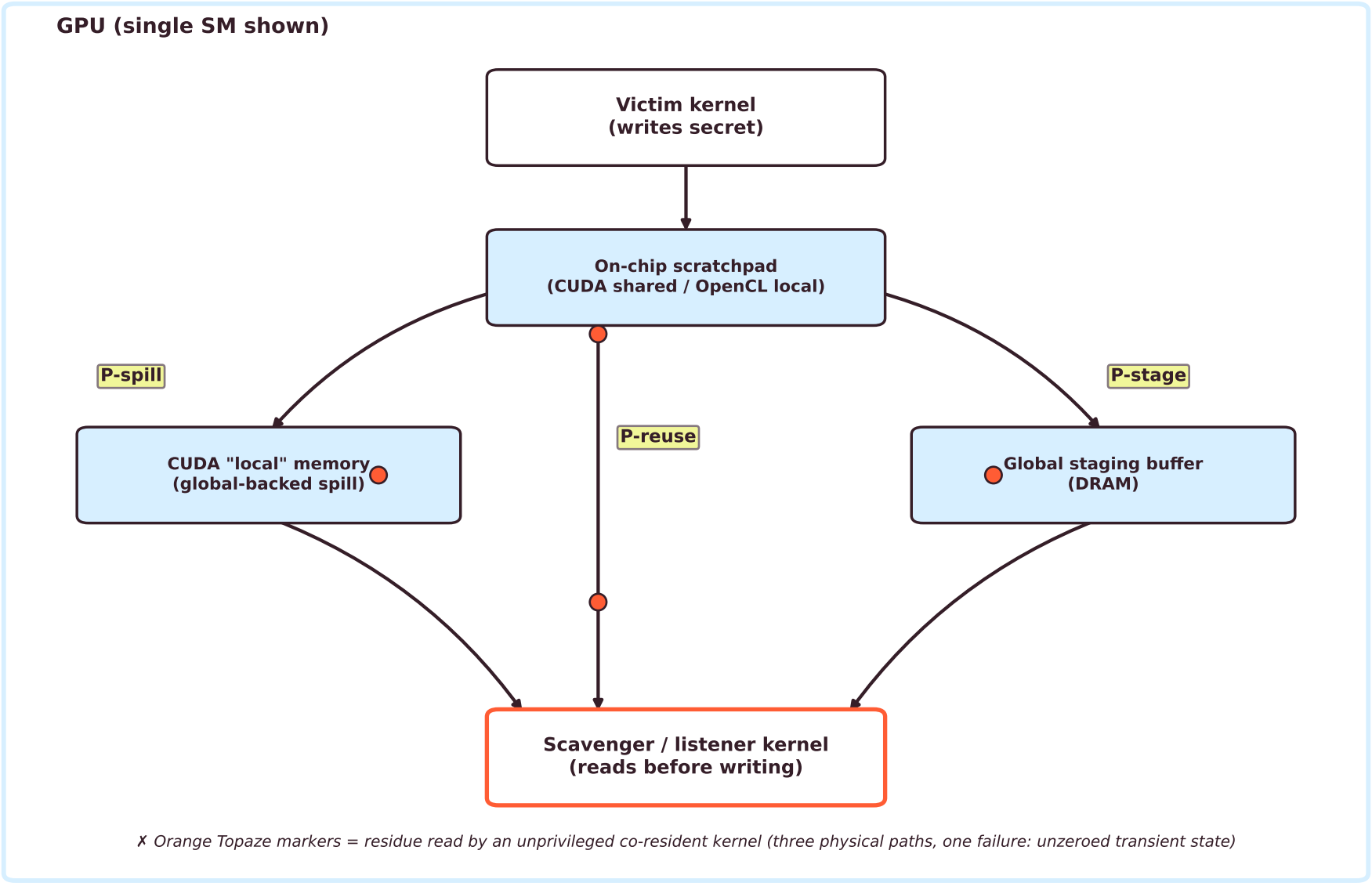Residue Lifecycle — Three Paths From One Failure
(P-reuse, P-spill, P-stage all terminate in a scavenger kernel reading unzeroed state)